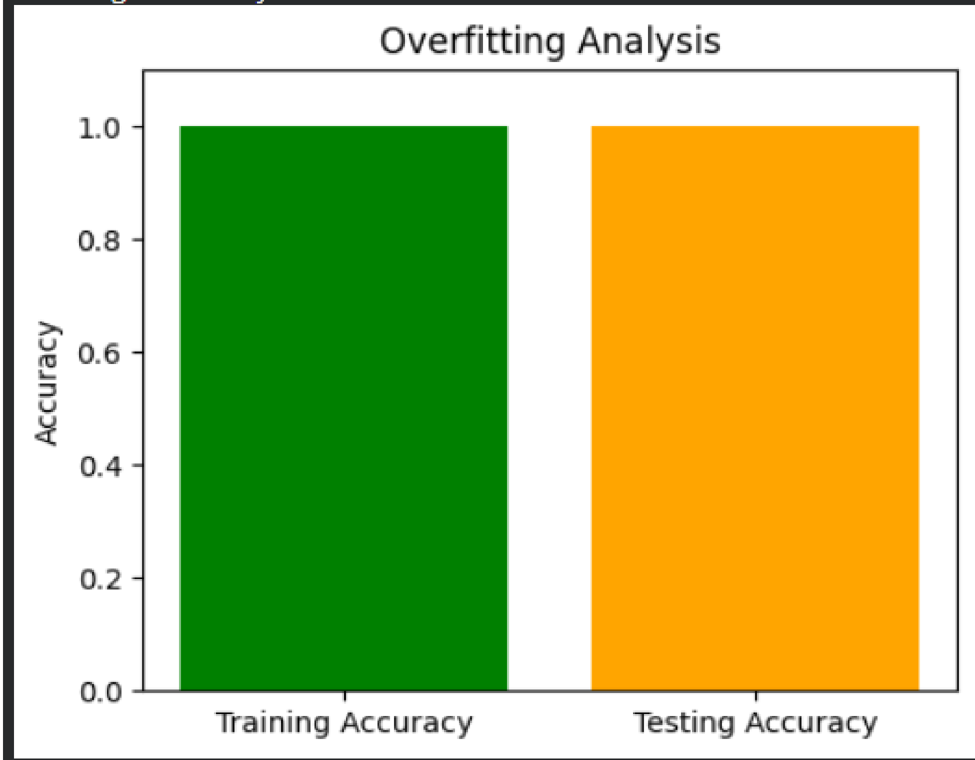
Experiment 4:

Aim: To analyse the performance of a over fitting by using choosen database with python code

Program:

```
# Import libraries
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score
import matplotlib.pyplot as plt

# Load Iris dataset
iris = load_iris()
X = iris.data
y = iris.target

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# Create a Decision Tree model (likely to overfit)
model = DecisionTreeClassifier(random_state=42)

# Train the model
model.fit(X_train, y_train)

# Predictions
train_pred = model.predict(X_train)
test_pred = model.predict(X_test)

# Calculate accuracy
train_accuracy = accuracy_score(y_train, train_pred)
test_accuracy = accuracy_score(y_test, test_pred)

# Print results
print("Training Accuracy :", train_accuracy)
print("Testing Accuracy  :", test_accuracy)

# Plot
plt.figure(figsize=(5,4))
plt.bar(["Training Accuracy", "Testing Accuracy"],
        [train_accuracy, test_accuracy],
        color=["green", "orange"])
plt.ylim(0, 1.1)
plt.title("Overfitting Analysis")
plt.ylabel("Accuracy")
```

```
Training Accuracy : 1.0
Testing Accuracy  : 1.0
```



plt.show()